Case Study: Stopping Cyberattacks at a Major International Airport

Estimated time needed: 5 minutes

Compromise

A major international airport uses an air-gapped network isolated from external access. This network manages various internal operations, from logistics to security. Despite this network isolation, numerous devices were discovered to be compromised by malware designed to capture and locally store data. This situation presented several security challenges: the airport, classified as critical infrastructure, could not afford any downtime; the network lacked comprehensive security safeguards; the isolated network linked devices with high and low-security levels; and there was no transparency regarding the activities within the network.

Discovery

Several behavioral anomalies were identified upon implementation of the IBM QRadar EDR platform. A malicious application was found to be installing an in-memory keylogger and a thread scrubbing the disk for sensitive files. The malware was extensive, containing mechanisms to bypass both local antivirus and sandbox analysis. The investigation uncovered two different vectors of infection, one installed at a public kiosk and another on a device that was part of the check-in management network sensor. Both vectors seemed to carry the same malware, attempting to contact the same command and control (C2) server.

Response

The QRadar EDR platform was critical in reconstructing the attack, even when deployed post-breach. The analysis showed that the infection occurred five months earlier from two separate USB drives. The malware spread to other endpoints mainly due to weak passwords and was able to collect information incessantly. However, the malware could not connect to the C2 server due to the network's isolation.

The malware demonstrated self-replication capabilities and could copy its storage to an external USB drive, although this function wasn't enabled. The malware was designed for manual exfiltration, suggesting the attackers intended to manually initiate this process. However, due to the network's isolation, this attempt was thwarted.

The response and remediation involved using the QRadar EDR remediation module to clear infected devices and storage folders. The threat-hunting interface was instrumental in confirming the absence of the malware vector from the entire infrastructure. The airport also established stricter internal traffic control rules, segregated the public from the operations network, and implemented continuous endpoint monitoring and regular threat-hunting campaigns.

Conclusion

This incident demonstrates that even with strong security measures like an air gap, vulnerabilities can still exist if not implemented strictly. The attackers' motives remain unclear as they collected data but never exfiltrated it. However, their open door into the airport's infrastructure could have led to significant operational disruptions. As such, robust and continuous security measures remain essential in protecting critical infrastructures like airports.

To read more about this case study: https://www.ibm.com/case-studies/major-international-airport

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